🎬 Smart Tamil Movie Recommendation System using Clustering

1. Introduction

Machine Learning is widely used to analyze patterns in data and provide intelligent recommendations. In the entertainment industry, recommendation systems help users discover movies that match their interests.

This project focuses on building a movie recommendation system for Tamil movies using clustering techniques. The system groups movies based on features such as rating, genre, duration, and popularity. When a user selects a movie, the system recommends similar movies from the same cluster.

The project demonstrates the use of unsupervised learning techniques from Machine Learning Unit IV.

2. Objective

The main objectives of this project are:

- To apply clustering techniques to group similar movies.
- To build an interactive movie recommendation system.
- To visualize movie clusters using graphs.
- To demonstrate practical use of machine learning algorithms.

3. Dataset Description

The dataset contains information about different Tamil movies.

Attributes used:

Attribute	Description
Movie	Name of the movie
Rating	Movie rating given by viewers
Genre	Category of the movie (Action, Comedy, Romance, etc.)
Duration	Length of the movie in minutes
Popularity	Popularity score based on audience interest

Example dataset entry:

Movie	Rating	Genre	Duration	Popularity
Vikram	8.3	Action	175	96
Leo	7.8	Action	164	95
Love Today	8.1	Romance	154	90

4. Methodology

The project follows the following steps:

1. Data Collection

Movie data is stored in a CSV file containing movie features.

2. Data Preprocessing

Categorical data such as genre is converted into numerical format using label encoding.

3. Feature Selection

Important features selected for clustering:

- Rating
- Duration
- Popularity
- Genre

4. Clustering

K-Means clustering is applied to group movies into clusters based on similarity.

5. Recommendation System

When a user selects a movie, the system recommends movies belonging to the same cluster.

6. Visualization

Graphs are used to visualize movie clusters based on rating and popularity.

5. Tools and Technologies Used

Tool	Purpose
Python	Programming language
Streamlit	Building the web application
Pandas	Data handling
Scikit-learn	Machine learning algorithms
Matplotlib	Data visualization

6. System Architecture

The workflow of the system is:

Movie Dataset

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Data Preprocessing

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Feature Selection

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K-Means Clustering

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User Selects Movie

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Recommendation System

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Display Results + Graph

7. Implementation

The application was implemented using Python and Streamlit. The movie dataset is loaded into the system, and clustering is performed using K-Means. The web interface allows users to select a movie and view its details along with recommended movies.

The system also generates a graph showing movie clusters based on rating and popularity.

8. Results

The system successfully groups movies into clusters and recommends similar movies based on user selection.

Example:

Selected Movie: **Vikram**

Recommended Movies:

- Leo
- Master
- Kaithi

The clustering graph shows how movies are grouped according to their characteristics.

9. Applications

This project can be used in:

- Movie streaming platforms
- Recommendation systems
- Entertainment applications
- Data analysis in media industries

10. Conclusion

This project demonstrates the application of machine learning clustering techniques for movie recommendation. Using K-Means clustering, movies were grouped based on their features, allowing the system to recommend similar movies effectively.

The project shows how machine learning can improve user experience by providing personalized recommendations.